Optimal Transport and Informed Trading in Securities Markets

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Contributions

1. Risk-averse market makers (normal asset values)
2. Risk-neutral market makers (general asset values)
3. Risk-averse market makers (general asset values)
4. Options and the underlying with risk-averse market makers

Risk aversion

- Market makers' marginal utility = SDF
- Like Grossman-Miller, Garleanu-Pedersen-Poteshman, ...
- Sort of a dynamic Grossman-Stiglitz model with a strategic informed trader
- Illiquidity depends on adverse selection and dealer aversion to inventory risk
- Mean-reverting prices, excess volatility, volatility risk premium, non-monotone SDF

Preview: Kyle's lambda with risk-averse dealers

One asset, $\tilde{v} \sim N(\mu, \sigma_v^2)$; noise-trade volatility σ_z; dealer risk aversion α; $\lambda_{\text{risk}} = \alpha\sigma_v^2$, $\lambda_{\text{info}} = \sigma_v/\sigma_z$

$$\mathrm{d}P_t = \lambda\,\mathrm{d}Y_t, \qquad \lambda = \frac{\lambda_{\text{risk}}}{2} + \sqrt{\lambda_{\text{info}}^2 + \left(\frac{\lambda_{\text{risk}}}{2}\right)^2}$$

Adverse selection only

$\alpha = 0$: $\lambda = \lambda_{\text{info}}$ (Kyle 1985)

Inventory risk only

$\sigma_z \to \infty$: $\lambda \to \lambda_{\text{risk}}$ (CARA–normal equilibrium)

Literature

- **Dealer inventory risk:** prices compensate dealers for holding unwanted inventory (Stoll 1978; Amihud and Mendelson 1980; Ho and Stoll 1981; Grossman and Miller 1988)
- **With adverse selection:** single period (Subrahmanyam 1991; Bollen, Smith and Whaley 2004); continuous-time Kyle (Çetin and Danilova 2016; Ying 2020)
- **Demand-based option pricing:** option dealers hedge in a frictionless market for the underlying, which has exogenous price dynamics (Gârleanu, Pedersen and Poteshman 2009)
- **This paper:** both markets are endogenous; dealers fill customer orders in the underlying and in the option

Roadmap

1. Risk-neutral market makers and optimal transport
2. Risk-averse market makers
3. Informed trading in a stock and a call option

Risk-neutral market makers and optimal transport

Model

- **Assets:** n risky assets with values $\tilde{v} \in \mathbb{R}^n$, distribution F, revealed at date 1; risk-free rate 0
- **Informed trader:** risk neutral, observes $\tilde{v}$ at date 0, trades at rate θ_t: $\mathrm{d}X_t = \theta_t \, \mathrm{d}t$
- **Noise traders:** cumulative orders Z, a Brownian motion with covariance matrix Σ_z
- **Market makers:** see only net order flow $Y = X + Z$; competitive and risk neutral:

$$P_t = \mathbb{E}\left[\tilde{v} \mid \mathcal{F}_t^Y\right]$$

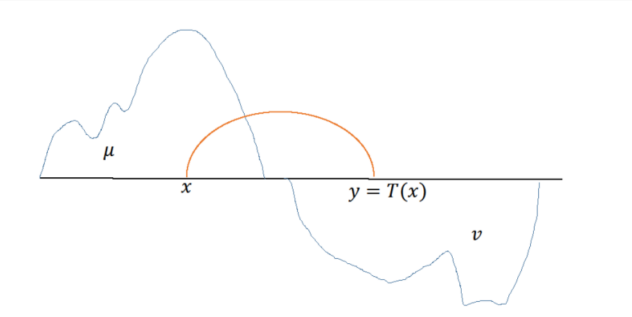
- **Insider's problem:** maximize $\mathbb{E}\left[\int_0^1 (\tilde{v} - P_t)' \theta_t \, \mathrm{d}t \;\middle|\; \tilde{v}\right]$ given the pricing rule $P_t = H(t, Y_t)$

From Kyle to a transport problem

- Look for an equilibrium in which Y_t is a sufficient statistic: $P_t = H(t, Y_t)$
- Insider hides in the noise: on $\mathcal{F}^Y$, the order flow Y is a $(0, \Sigma_z)$ Brownian motion on $\mathcal{F}^{\mathcal{Y}}$, so $Y_1 \sim N(0, \Sigma_z)$.
- No money left on the table: $P_1 = \tilde{v}$
- So $H(1, \cdot)$ maps $Y_1 \sim N(0, \Sigma_z)$ into $\tilde{v} \sim F$: it **transports** $N(0, \Sigma_z)$ to F

What is an optimal transport?

Move a pile of sand μ into a hole ν at least cost



- With quadratic cost, minimizing $\mathbb{E}|X - T(X)|^2$ is the same as maximizing $\mathbb{E}[X'T(X)]$; in one dimension the optimal T is monotone

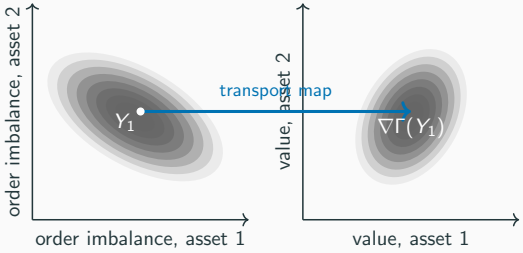
Kyle's lambda selects the map

- With n assets, $\mathrm{d}P_t = \Lambda_t \, \mathrm{d}Y_t$, and Λ_t must be symmetric and positive semidefinite
- $P_t = H(t, Y_t)$ implies $\Lambda_t = \nabla_y H(t, Y_t)$
- A symmetric positive semidefinite Jacobian means $H(t, \cdot)$ is the gradient of a convex function
- Brenier's theorem: exactly one transport of $N(0, \Sigma_z)$ to F is the gradient of a convex function

Brenier's theorem

Brenier (1991), McCann (1995)

For any distribution F on $\mathbb{R}^n$ there is a convex $\Gamma : \mathbb{R}^n \to \mathbb{R}$, unique up to a constant, such that $\nabla\Gamma(Z_1) \sim F$. If F is absolutely continuous, $\nabla\Gamma$ is invertible on the support of F and $\tilde{\zeta} := (\nabla\Gamma)^{-1}(\tilde{v}) \sim N(0, \Sigma_z)$.



Equilibrium with risk-neutral market makers

Normalize $\mathbb{E}[\Gamma(Z_1)] = 0$; insider strategies satisfy a no-doubling condition

Theorem 1

Let $K(t,y) = \mathbb{E}\big[\Gamma(y + Z_1 - Z_t)\big]$, the Gaussian smoothing of Γ. There is an equilibrium in which

(A) $P_t = \nabla_y K(t, Y_t)$ and $\mathrm{d}P_t = \Lambda(t, Y_t)\,\mathrm{d}Y_t$ with $\Lambda = \nabla_y^2 K$, symmetric positive semidefinite

(B) $P_t \to \tilde{v}$ as $t \to 1$

(C) the insider trades at rate $\theta_t = \dfrac{\tilde{\zeta} - Y_t}{1-t}$, where $\nabla\Gamma(\tilde{\zeta}) = \tilde{v}$

(D) Y is a $(0, \Sigma_z)$ Brownian motion on $\mathcal{F}^Y$ and $Y_1 = \tilde{\zeta}$

▸ Proof sketch

Why the insider targets $\tilde{\zeta}$

Under the pricing rule $P_t = \nabla_y K(t, Y_t)$, the insider's realized profit is

$$\int_0^1 (\tilde{v} - P_t)' \theta_t \, \mathrm{d}t \;=\; \underbrace{\tilde{v}' Y_1 - \Gamma(Y_1)}_{\text{depends only on } Y_1} \;-\; \underbrace{\int_0^1 (\tilde{v} - P_t)' \, \mathrm{d}Z_t}_{\text{mean zero}}$$

The dynamic problem reduces to a static choice of Y_1, and the optimum is $\tilde{v} = \nabla\Gamma(Y_1) \;\Leftrightarrow\; Y_1 \in (\nabla\Gamma)^{-1}(\tilde{v})$

The value of information

- Conditional on $\tilde{v} = v$, the insider's expected profit is the convex conjugate of the Brenier potential:

$$\Gamma^*(v) := \sup_y \{v'y - \Gamma(y)\}$$

- Example: one normal asset, $\tilde{v} \sim N(\mu, \sigma_v^2)$, $\lambda = \sigma_v/\sigma_z$:

$$\Gamma^*(v) = \frac{\lambda\sigma_z^2}{2} + \frac{(v-\mu)^2}{2\lambda}, \qquad \mathbb{E}\big[\Gamma^*(\tilde{v})\big] = \frac{\sigma_v\sigma_z}{2} + \frac{\sigma_v\sigma_z}{2} = \sigma_v\sigma_z$$

- New even for one asset with risk-neutral market makers

Monge–Kantorovich duality

$\Pi(F, G)$ = joint distributions of $(\tilde{v}, Y_1)$ with marginals F and $N(0, \Sigma_z)$

$$\sup_{\pi \in \Pi(F,G)} \mathbb{E}_\pi\big[\tilde{v}' Y_1\big] \;=\; \mathbb{E}\big[\Gamma^*(\tilde{v})\big] \;=\; \inf_{f(v)+g(y) \geq v'y} \mathbb{E}\big[f(\tilde{v})\big] + \mathbb{E}\big[g(Y_1)\big]$$

- **Primal:** as if the insider chooses Y_1 as a function of $\tilde{v}$ to maximize $\mathbb{E}[\tilde{v}' Y_1]$, subject to Y_1 having the noise distribution $N(0, \Sigma_z)$
- **Dual:** profit bounded by a separable $f(v) + g(y)$; the minimizers are $f = \Gamma^*$ and $g = \Gamma$

Kyle objects are optimal-transport objects

Risk-neutral market makers, n assets, any distribution F

Kyle model	**Optimal transport**
Terminal pricing rule $H(1,\cdot)$	Brenier map $\nabla\Gamma$
Insider's target Y_1	Inverse map: $Y_1 \in (\nabla\Gamma)^{-1}(\tilde{v})$
Kyle's lambda Λ	Hessian of the smoothed potential, $\nabla_y^2 K$
Conditional expected profit	Convex conjugate $\Gamma^*(\tilde{v})$
Unconditional expected profit	Wasserstein distance $W_2(F,G)$

Market-maker profits and transaction costs

- Realized market-maker profits, with the optimal insider strategy:

$$\langle P, Y\rangle_1 - \Gamma^*(\tilde{v}), \qquad \langle P, Y\rangle_1 = \operatorname{tr}\left(\Sigma_z \int_0^1 \Lambda(t, Y_t)\,\mathrm{d}t\right)$$

- $\langle P, Y\rangle_1$ is price-impact revenue from noise traders ($\sum_i \mathrm{d}P_i\,\mathrm{d}Y_i$)
- The insider pays no such cost: her orders are of order $\mathrm{d}t$; noise orders are of order $\sqrt{\mathrm{d}t}$

Risk-averse market makers

Two ways to price with risk-averse market makers

- **Utility-indifference pricing:** dealers are indifferent about trading at all (Subrahmanyam 1991; Çetin and Danilova 2016)
- **Walrasian pricing (this paper):** dealers are indifferent about *further trades at the margin* (Stoll 1978; Grossman and Miller 1988; Gârleanu, Pedersen and Poteshman 2009; He and Krishnamurthy 2013)
- Interpretation: a competitive inter-dealer market (Naik, Neuberger and Viswanathan 1999), or a competitive fringe of uninformed investors

A two-dealer example

Each dealer has utility $u(x) + y$ for x shares and y cash, holds n shares; a market order to buy 2 shares arrives

- **No inter-dealer market:** each dealer sells one share at $p^\dagger = u(n) - u(n-1)$, which leaves her utility unchanged
- **Inter-dealer market:** it clears at $p^* = u'(n-1)$; Bertrand competition drives the customer price to p^*, the marginal cost of buying in the inter-dealer market
- u concave $\Rightarrow p^* > p^\dagger$: the dealers' gain comes from access to a competitive market

Equilibrium conditions

- CARA market makers aggregate to a representative market maker with risk aversion α. For simplicity, take initial inventory $\beta = 0$.
- Dealer wealth at date 1: $\tilde{w} = \langle P, Y \rangle_1 - \int_0^1 (\tilde{v} - P_t)' \, \mathrm{d}Y_t$
- SDF and risk-neutral probability: $\dfrac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = \tilde{m} = \dfrac{\mathrm{e}^{-\alpha \tilde{w}}}{\mathbb{E}[\mathrm{e}^{-\alpha \tilde{w}}]}$
- Prices: $P_t = \mathbb{E}^{\mathbb{Q}}\left[\tilde{v} \mid \mathcal{F}_t^Y\right]$; the insider stays risk neutral under $\mathbb{P}$

Three conditions: (1) the insider optimizes; (2) prices are $\mathbb{Q}$-expectations; (3) $\mathbb{Q}$ is consistent with equilibrium dealer wealth.

Solution method

- Start with a risk-neutral marginal distribution F^* for $\tilde{v}$. Prices should be risk-neutral expected values, so use Brenier potential for F^*. Define prices using it.
- Insider maximizes conditional on $\tilde{v}$, so its distribution does not matter to her. She regards Z as a $(0, \Sigma_z)$–Brownian motion. Expected profits are again maximized by steering Y_1 to $(\nabla\Gamma)^{-1}(\tilde{v})$.
- The risk-neutral pricing condition follows as before.

What is the physical distribution?

- Realized market-maker profits are again

$$\tilde{w} = -Y_1'\tilde{v} + \Gamma(Y_1) + \operatorname{tr}\Big(\Sigma_z \int_0^1 \nabla_y^2 K(t, Y_t)\,\mathrm{d}t\Big)$$

- We know we should have $\mathrm{d}\mathbb{P}/\mathrm{d}\mathbb{Q} \propto \mathrm{e}^{\alpha\tilde{w}}$
- To avoid circularity, define F to be the distribution of $\nabla\Gamma(Z_1)$ under a measure change of Z from $(0, \Sigma_z)$–Brownian motion using the Radon-Nikodym derivative

$$\tilde{\chi} := \exp\Big\{ -\alpha Z_1'\nabla\Gamma(Z_1) + \alpha\Gamma(Z_1) + \alpha \operatorname{tr}\Big(\Sigma_z \int_0^1 \nabla_z^2 K(t, Z_t)\,\mathrm{d}t\Big)\Big\}$$

- Actually, the RN derivative is $\tilde{\chi}/\mathbb{E}[\tilde{\chi}]$ and we need $\mathbb{E}[\tilde{\chi}] < \infty$.

What F's can we get like this

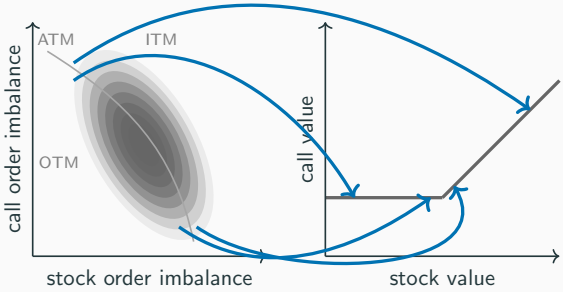
All normal F's, all F's with bounded support, and more.

Informed trading in a stock and a call option

Setup

- A stock and a European call with strike 100, maturing at date 1 (Back 1993, now with risk-averse dealers)
- Risk-neutral stock value: $\log \tilde{v}_s = \mu + \sigma\tilde{x}$, $\tilde{x}$ standard normal truncated to $|\tilde{x}| \leq 3.3$; $\sigma = 0.2$, $\mu = \log 100 - \sigma^2/2$
- $\alpha = 0.2$; four noise covariance matrices Σ_z
- 10,000 simulated paths for each Σ_z
- Figures: $\Sigma_z = \left(\begin{smallmatrix} 4 & -2 \\ -2 & 4 \end{smallmatrix}\right)$ unless noted

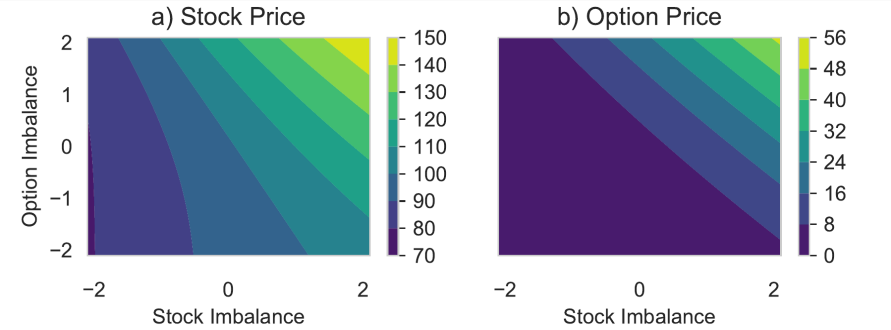
Transport onto the option payoff



- The target $(v_s, (v_s - 100)^+)$ lies on a curve, so $\nabla\Gamma$ is not invertible: the insider randomizes over $(\nabla\Gamma)^{-1}(\tilde{v})$ so that $Y_1 \sim N(0, \Sigma_z)$

Prices

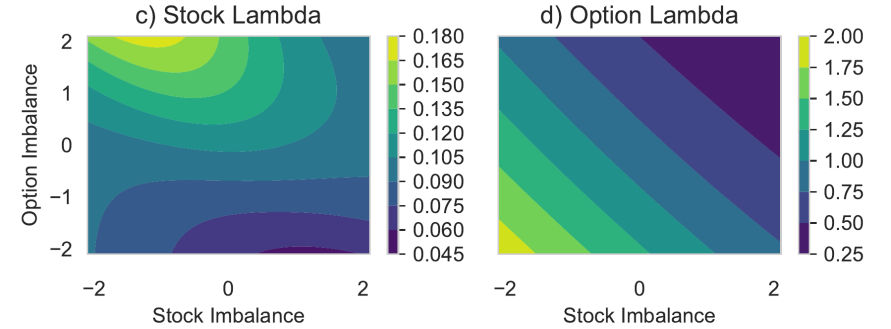
$t = 0.5$; state variables are the stock and option order imbalances



- In the money (top right), both prices depend on roughly $y_s + y_o$; deep out of the money, the stock price ignores option orders

Price impact

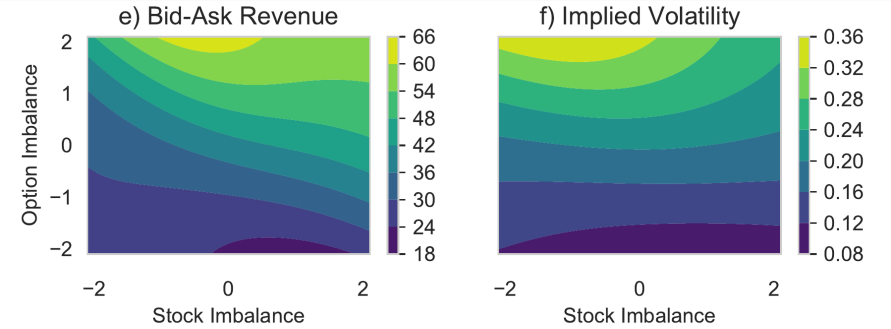
$t = 0.5$; lambdas are diagonal elements of Λ divided by price



- The option lambda is far larger than the stock lambda; the stock lambda is highest where the stock was sold and the call bought

Dealer revenue and implied volatility

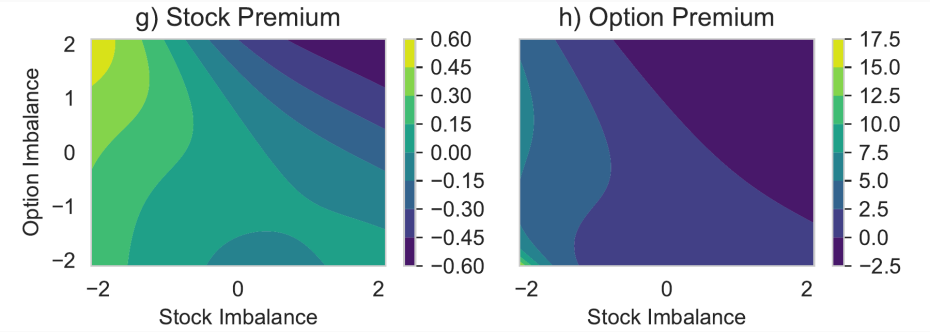
$t = 0.5$; revenue is $\mathrm{tr}(\Sigma_z \nabla_y^2 K)$ per unit time; Black–Scholes implied volatility



- Both are high in the top left: stock sold, call bought, and dealers most uncertain about the stock value

Risk premia

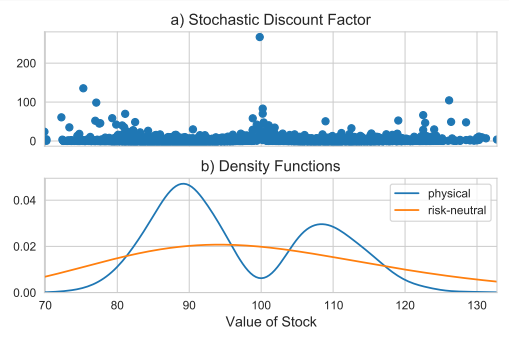
$t = 0.5$; expected rates of return per unit time



- The stock premium is highest in the top left, where implied volatility is also high

The SDF and the physical distribution

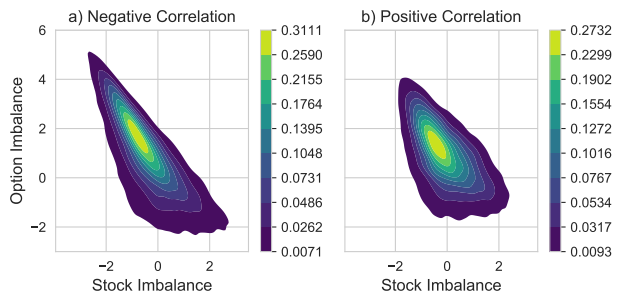
10,000 simulations; lognormal risk-neutral density, kernel estimate of the physical density



- The SDF is high for extreme stock values and near the strike, so the risk-neutral distribution has fatter tails than the physical one

Dealers end up hedged

Physical density of terminal order imbalances (Y_s, Y_o); 10,000 simulations each



- Net imbalances are negatively correlated for either sign of noise correlation: dealers are short calls and long stock (as in Ni, Pearson, Poteshman and White 2021)

Implied volatility predicts stock returns

Stock risk premium on change in implied volatility, $t = 0.5$; 10,000 simulations per column; standard errors in parentheses

Σ_z	$\begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}$	$\begin{pmatrix} 4 & -1 \\ -1 & 1 \end{pmatrix}$	$\begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}$	$\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$
Δ implied vol	0.444	0.354	0.381	0.253
	(0.030)	(0.039)	(0.052)	(0.060)
R^2	0.021	0.008	0.005	0.002

- Positive in all four cases, as An, Ang, Bali and Cakici (2014) find in the data. With risk-neutral dealers the premium is zero. Adding the prior stock return flips the sign when noise trades are positively correlated

▸ Full table

Conclusion

- The dynamic Kyle equilibrium is an optimal transport: the Brenier map gives prices, Kyle's lambda and the insider's target; its convex conjugate gives her profits
- With risk-averse dealers, transport to the risk-neutral distribution; illiquidity combines adverse selection and inventory risk
- Maybe some interesting empirical predictions about options

Appendix

Proof sketch of Theorem 1

- $K(t,y) = \mathbb{E}[\Gamma(y + Z_1 - Z_t)]$ solves the backward heat equation $K_t + \frac{1}{2}\operatorname{tr}\left(\Sigma_z \nabla_y^2 K\right) = 0, \;\; K(1,\cdot) = \Gamma$
- So $P = \nabla_y K(t, Y)$ is a martingale whenever Y is a $(0, \Sigma_z)$ Brownian motion
- Candidate value function: $J(t,v,y) = \Gamma^*(v) + K(t,y) - y'v$
- $J_y = \nabla_y K - v = -(v - P)$: the gain from trading faster is exactly offset by the loss in J, so every admissible strategy with $Y_1 = \tilde{\zeta}$ is optimal
- The bridge strategy hits $\tilde{\zeta}$, and $\tilde{\zeta} \sim N(0, \Sigma_z)$, so Y remains a Brownian motion on $\mathcal{F}^Y$

Back

Noise-trader profits

- From the Merton budget constraint and integration by parts, noise traders earn

$$\int_0^1 (\tilde{v} - P_t)' \, \mathrm{d}Z_t - \langle P, Z \rangle_1$$

- The same formula for the insider has $\langle P, X \rangle = 0$ because $\mathrm{d}X = \theta \, \mathrm{d}t$
- Market makers earn the negative of the sum:

$$\langle P, Y \rangle_1 - \int_0^1 (\tilde{v} - P_t)' \, \mathrm{d}Y_t$$

Related literature

- **Continuous-time Kyle:** Kyle (1985); Back (1992); Back (1993), stock and call with a parametric assumption; Lasserre (2004), Gaussian-homeomorphic multi-asset; Back and Crotty (2015)
- **Optimal transport and informed trading:** Kramkov and Xu (2022); Malamud and Schrimpf (2021); both single-period
- **Risk-averse market makers:** Çetin and Danilova (2016), utility indifference; Ying (2020), representative investor over the aggregate dividend
- **Later work:** Çetin and Danilova (2026) recover our one-asset normal result; Çetin (2025), entropy-regularized transport with transaction costs
- **Intermediary asset pricing:** He and Krishnamurthy (2013); Adrian, Etula and Muir (2014); Haddad and Muir (2021)

Many normal assets: formulas

$\tilde{v} \sim N(\mu, \Sigma_v)$; $\Sigma_z^{1/2}\Sigma_v\Sigma_z^{1/2} = VEV'$, $E = \mathrm{diag}(e_i)$

$$\sqrt{d_i} = \frac{\alpha e_i}{2} + \sqrt{\frac{\alpha^2 e_i^2}{4} + e_i}, \qquad D = \mathrm{diag}(d_i)$$

$$\Lambda = \Sigma_z^{-1/2} V D^{1/2} V' \Sigma_z^{-1/2}, \qquad \Sigma_v^* = \Sigma_z^{-1/2} V D V' \Sigma_z^{-1/2}, \qquad \mu^* = \mu - \alpha \Sigma_v \beta$$

$$\mathrm{d}Y_t = \Delta_t(\beta - Y_t)\,\mathrm{d}t + \Sigma_z^{1/2}\,\mathrm{d}B_t, \qquad \Delta_t = \Sigma_z^{1/2} V C_t V' \Sigma_z^{-1/2}, \qquad c_{it} = \frac{\alpha\sqrt{d_i}}{1 + (1-t)\alpha\sqrt{d_i}}$$

Back

General model: the change of measure

$\nabla\Gamma$ transports $N(0,\Sigma_z)$ to F^*, $\mathbb{E}[\Gamma(Z_1)]=0$, $\nu=$ Wiener measure of Z

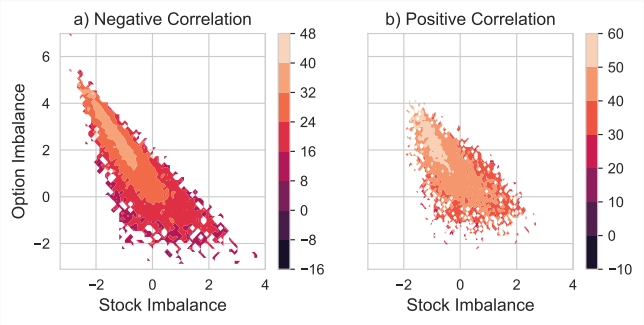
$$\tilde{\chi}=\exp\Big\{-\alpha(Z_1-\beta)'\nabla\Gamma(Z_1)+\alpha\Gamma(Z_1)+\alpha\,\mathrm{tr}\left(\Sigma_z\int_0^1\nabla_z^2K(t,Z_t)\,\mathrm{d}t\right)\Big\}$$

- The physical distribution F is the distribution of $\nabla\Gamma(Z_1)$ under $\mathrm{d}\hat{\nu}/\mathrm{d}\nu=\tilde{\chi}/\mathbb{E}^{\nu}[\tilde{\chi}]$
- Normal F^* gives normal F, and every normal F arises this way
- Under $\mathbb{P}$: $\mathrm{d}Y_t=\Sigma_z\nabla_y\phi(t,Y_t)\,\mathrm{d}t+\Sigma_z^{1/2}\,\mathrm{d}B_t$ (Girsanov drift)
- Insider: $\theta_t=\dfrac{\tilde{\zeta}-Y_t}{1-t}+\Sigma_z\nabla_y\psi(t,Y_t,\tilde{\zeta})$

Back

Market-maker profits by terminal imbalance

Realized profits averaged within bins of a 100×100 grid; 10,000 simulations each



- Profits are highest after stock sales and call purchases, where dealers are most uncertain and price impacts are highest

Implied volatility and the prior return

Stock risk premium on Δ implied volatility and prior stock return, $t = 0.5$; standard errors in parentheses

Σ_z	$\left(\begin{smallmatrix} 4 & -2 \\ -2 & 4 \end{smallmatrix}\right)$	$\left(\begin{smallmatrix} 4 & -1 \\ -1 & 1 \end{smallmatrix}\right)$	$\left(\begin{smallmatrix} 4 & 2 \\ 2 & 4 \end{smallmatrix}\right)$	$\left(\begin{smallmatrix} 4 & 1 \\ 1 & 4 \end{smallmatrix}\right)$
Δ implied vol	0.491	0.547	−0.603	−0.188
	(0.010)	(0.010)	(0.014)	(0.014)
Prior stock return	−2.187	−1.774	−2.360	−1.830
	(0.008)	(0.005)	(0.007)	(0.004)
R^2	0.894	0.932	0.928	0.943

- The prior-return coefficient is always negative (reversals); the implied-volatility sign follows the correlation between the two regressors

Back

Numerical method

- **Transport map:** linear programming between a 151×151 grid of order imbalances and ~ 1001 points on the payoff curve (POT library); chosen because it handles the discontinuities in $\nabla\Gamma$
- **Potential:** a feed-forward network fit so its gradient matches $\nabla\Gamma$, with a convexity penalty
- **Backward recursion:** K, $\nabla_y K$ and e^{ϕ} by dynamic programming, $\Delta t \approx 1/100$
- **Physical distribution:** Euler simulation of Y under $\mathbb{P}$; $\nabla\Gamma(Y_1)$ gives F